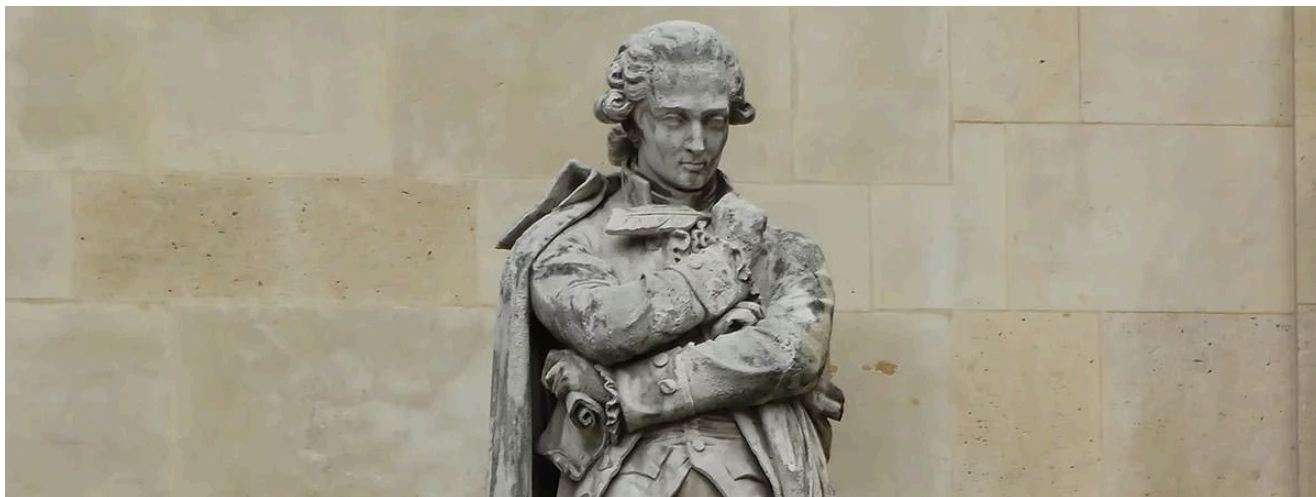




10 ВАЖНЕЙШИХ ОТКРЫТИЙ АНТУАНА ЛАВУАЗЬЕ

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Антуан Лоран де Лавуазье (1743–1794) был французским химиком, наиболее известным тем, что превратил химию из качественной науки в количественную и открыл роль кислорода в процессе горения. До Лавуазье господствующей теорией горения была теория флогистона, которая в конечном итоге была опровергнута его работами. Лавуазье сделал множество других важных открытий в области химии, в том числе установил, что вода — это соединение водорода и кислорода; доказал, что сера — это химический элемент, а алмаз — это форма углерода; сформулировал закон сохранения массы в химии и стал соавтором первой современной системы химической номенклатуры. Лавуазье оказал огромное

влияние на историю химии и известен как «отец современной химии».

Узнайте больше об изобретениях, открытиях и других достижениях Антуана Лавуазье, ознакомившись с его 10 главными открытиями.

#1 ОН ОПРОВЕРГ РАСПРОСТРАНЕННУЮ ТЕОРИЮ ФЛОГИСТОНА О ГОРЕНИИ

В начале XVIII века немецкий учёный *Георг Эрнст Шталь* предложил теорию флогистона для объяснения процесса горения, которая получила широкое распространение. Согласно этой теории, *каждое горючее вещество содержит универсальный компонент огня, называемый флогистоном*. Это вещество выделяется при горении, дыхании и прокаливании и поглощается при обратных процессах. Однако при нагревании металлов образующийся оксид *весил больше, чем исходный металл*. Но, согласно гипотезе Шталя, он должен был весить меньше, поскольку металл *терял флогистиновый компонент*. Сторонники этой теории даже предполагали, что флогистон может иметь отрицательный вес. В 1772 году Антуан Лавуазье провел свои первые эксперименты по изучению горения. Он сообщил, что при сжигании фосфора и серы *они увеличивали вес, соединяясь с воздухом* и что *продукты горения были кислотными*. Лавуазье занимался изучением горения в течение следующих пятнадцати лет, и его работа *в конечном итоге опровергла теорию флогистона о горении*.

Портрет Антуана Лавуазье

#2 ОН ВЫДВИНУЛ КИСЛОРОДНУЮ ТЕОРИЮ ГОРЕНИЯ

Антуан Лавуазье понял, что элементы вступают в реакцию с чем-то в воздухе, в результате чего их вес увеличивается. В 1774 году английский учёный *Джозеф Пристли* выделил компонент воздуха, нагревая ртутную окалину (оксид ртути). Он считал, что это *чистая форма воздуха*, поскольку она способствует дыханию и горению. Пристли назвал его *дефлогистированным воздухом*, полагая, что его необычные свойства обусловлены отсутствием флогистона. Когда он сообщил Лавуазье о своем открытии, тот повторил эксперимент с ртутью и другими оксидами металлов. Он пришел к выводу, что воздух состоит из двух компонентов: один из них вступает в реакцию с металлом и поддерживает дыхание, а другой не поддерживает ни горения, ни дыхания. В 1778 году Лавуазье

выдвинул свою новую теорию горения, согласно которой *горение* — это реакция металла или органического вещества с той частью обычного воздуха, которую он назвал «наиболее пригодной для дыхания». В следующем году он придумал для нее название «кислород», образовав его от греческих слов, означающих «порождающий кислоту». Открытие Лавуазье роли кислорода в процессе горения считается одним из его главных достижений.

Comparison of phlogiston theory and Lavoisier's theory

#3 HE CO-ESTABLISHED THAT WATER WAS A COMPOUND AND NOT AN ELEMENT

In 1783, Antoine Lavoisier coined the name “*hydrogen*” for the gas which *Henry Cavendish* had recognized as a new element in 1766. Cavendish had called the gas *inflammable air*. In cooperation with French mathematician *Pierre Simon de*

Laplace, Lavoisier began a series of experiments on the composition of water in 1783. The two burned jets of hydrogen and oxygen in a bell jar over mercury to obtain *“water in a very pure state”*. The quantitative results were good enough to support the contention that *water was not an element but a compound of two gases, hydrogen and oxygen*. This was a remarkable discovery as *everyone had considered water to be an element* from the time of Aristotle who included it in his four elements; over 2,000 years ago. The interpretation of water as compound also explained the inflammable air (hydrogen) generated from dissolving metals in acids and the reduction of oxides by the inflammable air.

Pierre-Simon Laplace – Who worked with Lavoisier to
establish that water was a compound

#4 HE CO-DISCOVERED THAT RESPIRATION IS A FORM OF COMBUSTION

In 1782–83, along with Pierre Simon de Laplace, Lavoisier conducted experiments in the area of *respiration physiology*. The pair used a calorimeter to measure the amount of heat given off by a guinea pig in a measured interval of time. They also measured the amount of carbon dioxide (then called fixed air) given off by the guinea pig in this same interval. They found that a similar amount of heat was produced when sufficient carbon was burned in the ice calorimeter to produce the same amount of carbon dioxide as that which the guinea pig exhaled. From this, Lavoisier and Laplace concluded that *respiration was similar to slow combustion*. This enables the living animal to maintain its body temperature above that of its surroundings. The experiment *accounted for the puzzling phenomenon of animal heat*. The pioneering work of Lavoisier and Laplace in the field *served to inspire similar research on physiological processes for generations to come*.

Depiction of Antoine Lavoisier conducting an experiment on respiration in the 1770s

#5 HE CO-AUTHORED THE FIRST MODERN SYSTEM OF CHEMICAL NOMENCLATURE

Together with French chemists *Louis-Bernard Guyton*, *Claude Louis Berthollet* and *Antoine Francois*, Lavoisier published in 1787 a work titled *Méthode de nomenclature chimique* (*Method of Chemical Nomenclature*). This was the *first proper system of chemical nomenclature*, i.e. a system of names describing the structure of chemical compounds. It was based on three general principles: substances should have one fixed name; it should reflect composition when known; and it should generally be chosen from Greek or Latin roots. 55 substances which could not be decomposed into simpler substances by any known chemical means were listed as *elements* in the publication. The list was not totally accurate and included *light* and *caloric* (*matter of heat*). The acids, which were recognized as compounds in the system, were given names according to the degree of oxygenation, like *nitric* and *nitrous* acids. The “ic”

termination indicated acids with a higher proportion of oxygen than those with the “ous” ending. The new nomenclature *spread throughout the world and became common use in the field of chemistry.*

Part of the list of elements in Lavoisier's Elementary Treatise of
Chemistry

#6 HE IS CREDITED WITH ESTABLISHING MASS CONSERVATION IN CHEMICAL REACTIONS

In 1778, Lavoisier found that when mercury oxide is heated its weight decreases; and the oxygen released *has the same weight as the weight lost by mercury oxide*. After carrying out work with a number of different substances, he concluded that this was due to the *law of conservation of mass*, which states that *the total mass of matter is the same at the end as at the beginning of every chemical change*. Though the principle of conservation of matter had been stated by several people earlier, Lavoisier illustrated it with experiments and employed a criteria for conservation: *the total mass of the products must come from the mass of the reactants*. The law of conservation of mass became established only after Lavoisier's efforts and many credit him for *discovering mass conservation in chemical reactions*. In fact in France, the law is still taught as *Lavoisier's Law*.

Diagram explaining Law of Conservation of Mass

#7 HE DISCOVERED THAT SULFUR IS AN ELEMENT AND THAT DIAMOND IS A FORM OF CARBON

In 1772, Antoine Lavoisier and other chemists placed a diamond in a glass jar and focused sun's rays on it with a giant magnifying glass. The diamond burned and disappeared. Lavoisier found that whether diamond or charcoal was burnt, neither produced any water and both released the same amount of carbon dioxide per gram. He thus discovered that *diamond is a crystalline form of carbon* introducing the possibility of allotropy in chemical elements. In 1777, Lavoisier carried out extensive experiments involving sulfur and found that it could not be broken down into any simpler substances. He thus became *the first person to establish that sulfur was an element and not a compound*. In 1787, Lavoisier suspected that silica might be an oxide of a fundamental chemical element thus *predicting the existence of silicon*.

#8 HE WROTE THE FIRST MODERN TEXTBOOK ON CHEMISTRY

In 1789, Antoine Lavoisier published his most famous work *Traité élémentaire de chimie* (*Elementary Treatise of Chemistry*). The book established Lavoisier's oxygen theory of combustion and denied the existence of phlogiston. It defined an element as *a single substance that can't be broken down by chemical analysis and from which all chemical compounds are formed*. It contained a list of elements, which formed the basis for the modern list of elements. It explained the influence of heat on chemical reactions; the nature of gases; the reactions of acids and bases to form salts; and the apparatus used to perform chemical experiments. It also presented a unified view of new theories of chemistry and contained a clear statement of the law of conservation of mass. Elementary

Treatise is regarded as the *first modern textbook on the subject of Chemistry*. It went on to be *hugely influential* and remains a *classic in the history of science*.

Front page of Elementary Treatise of Chemistry by
Antoine Lavoisier

#9 LAVOISIER IS CONSIDERED THE FATHER OF MODERN CHEMISTRY

Lavoisier is most famous for *changing chemistry from a qualitative to a quantitative science*. He performed *some of the first truly quantitative chemical experiments*. He carefully weighed the reactants and products of a chemical reaction in a sealed glass vessel so that no gases could escape, which was a crucial step in the advancement of chemistry. Lavoisier is considered a *pioneer of stoichiometry*, branch of chemistry concerned with calculation of relative quantities of reactants and products in chemical reactions. He was *responsible*

for the construction of the gasometer, a large container in which natural gas is stored. It enabled him to weigh the gas in a pneumatic trough with the precision he required. He also established the consistent use of the chemical balance, a device used to measure weight. The work of Lavoisier raised the level of chemistry leading to it becoming as important as physics and mathematics. For all his accomplishments in the field, Antoine Lavoisier is widely regarded as the "father of modern chemistry".

Statue of Antoine Lavoisier at the Louvre, Paris

#10 HE CONTRIBUTED IN THE ADOPTION OF THE METRIC SYSTEM

Apart from his contributions to science, Antoine Lavoisier also did a lot of work as a humanitarian. In 1765, he submitted an essay on improving urban street lighting to the *French Academy of Sciences* for which he was awarded a gold

medal by King Louis XV. He worked on projects to purify the water from the Seine; to improve air quality and study health risks associated with gunpowder's effect on the air; to improve living conditions of prisoners; to reform the French monetary and taxation system to help the peasants; and to improve the agricultural yields in the Sologne. In 1791, Lavoisier *chaired the commission set up to establish a uniform metric system.* Other members of the committee including the well-known mathematicians Pierre-Simon Laplace and Adrien-Marie Legendre. The new system of uniform weights and measures was adopted by the Convention on *1 August 1793.*

📁 Accomplishments, Scientists

🔖 Famous French, Science

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